DSA4213 Final Project

# NLP-Powered Course Recommender: **EduQuest**

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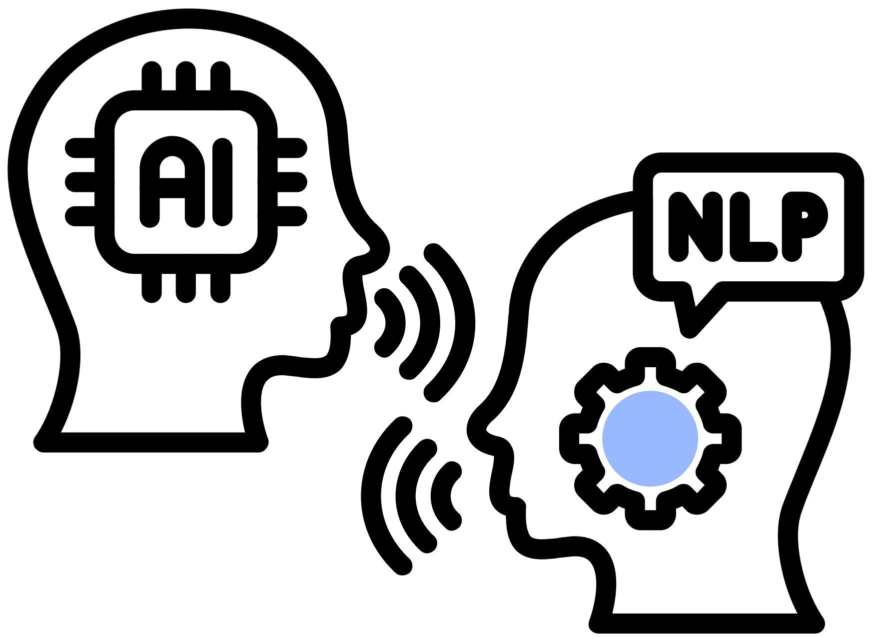


# MOTIVATION

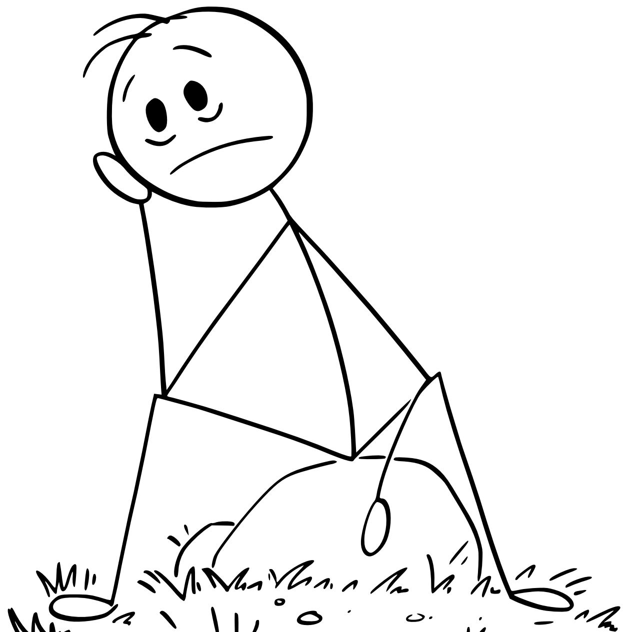
Students find it hard to pick courses that

Match their interests

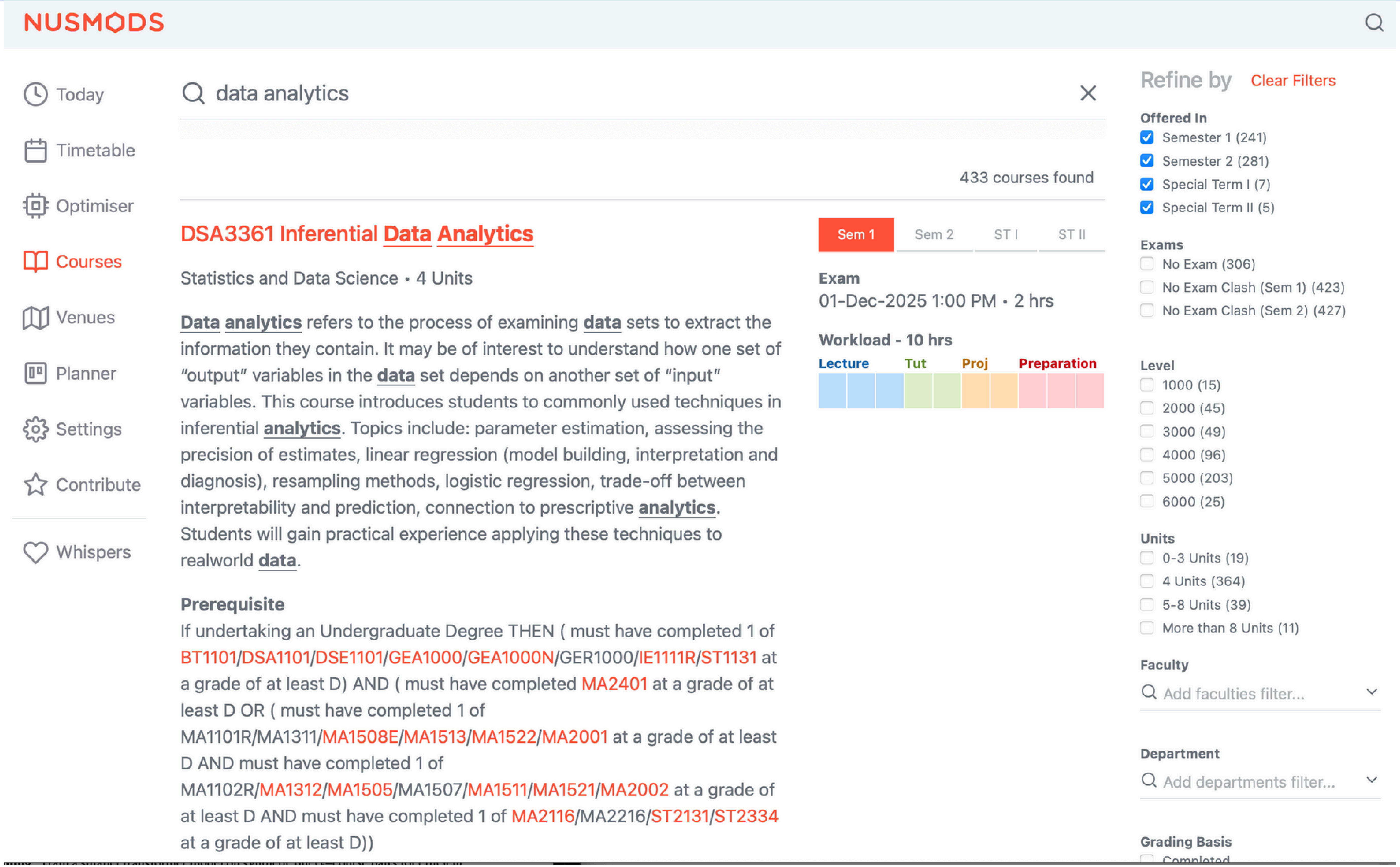
# PROBLEM

NUSMods search focuses on keywords, not student intent

Not personalized reasoning



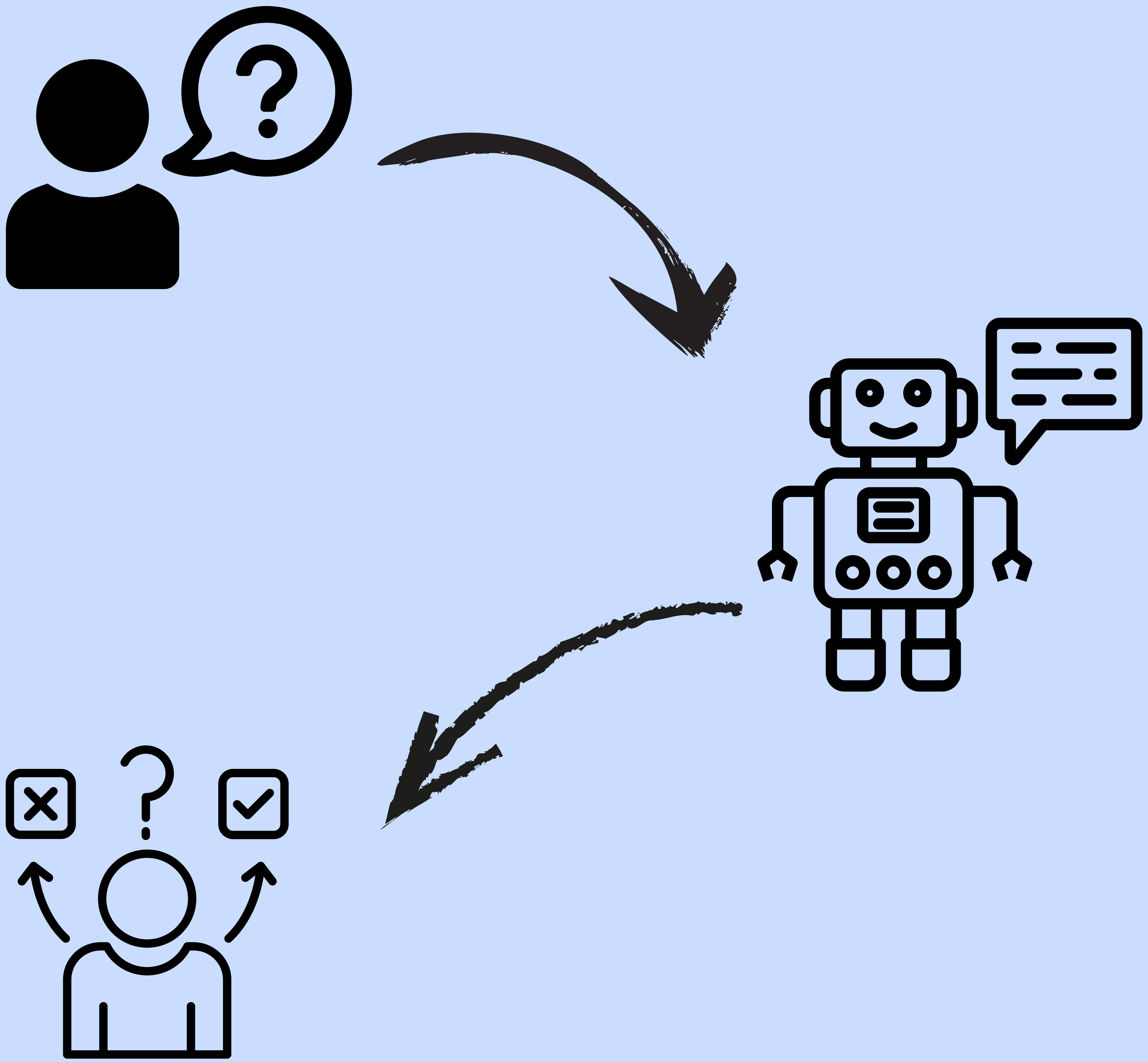
# Current NUSMods: Keyword Search, Not Intelligent Discovery



**Could we let an LLM decide?**

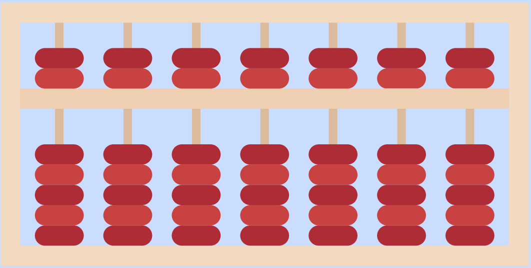
**not by keywords**

but by **understanding what the student really wants?**



# TRADITIONAL QUERY

I want to learn machine learning and deep learning for real-world applications.

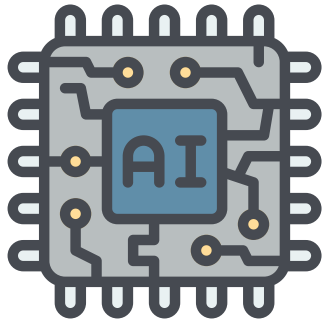


# ENHANCED QUERY

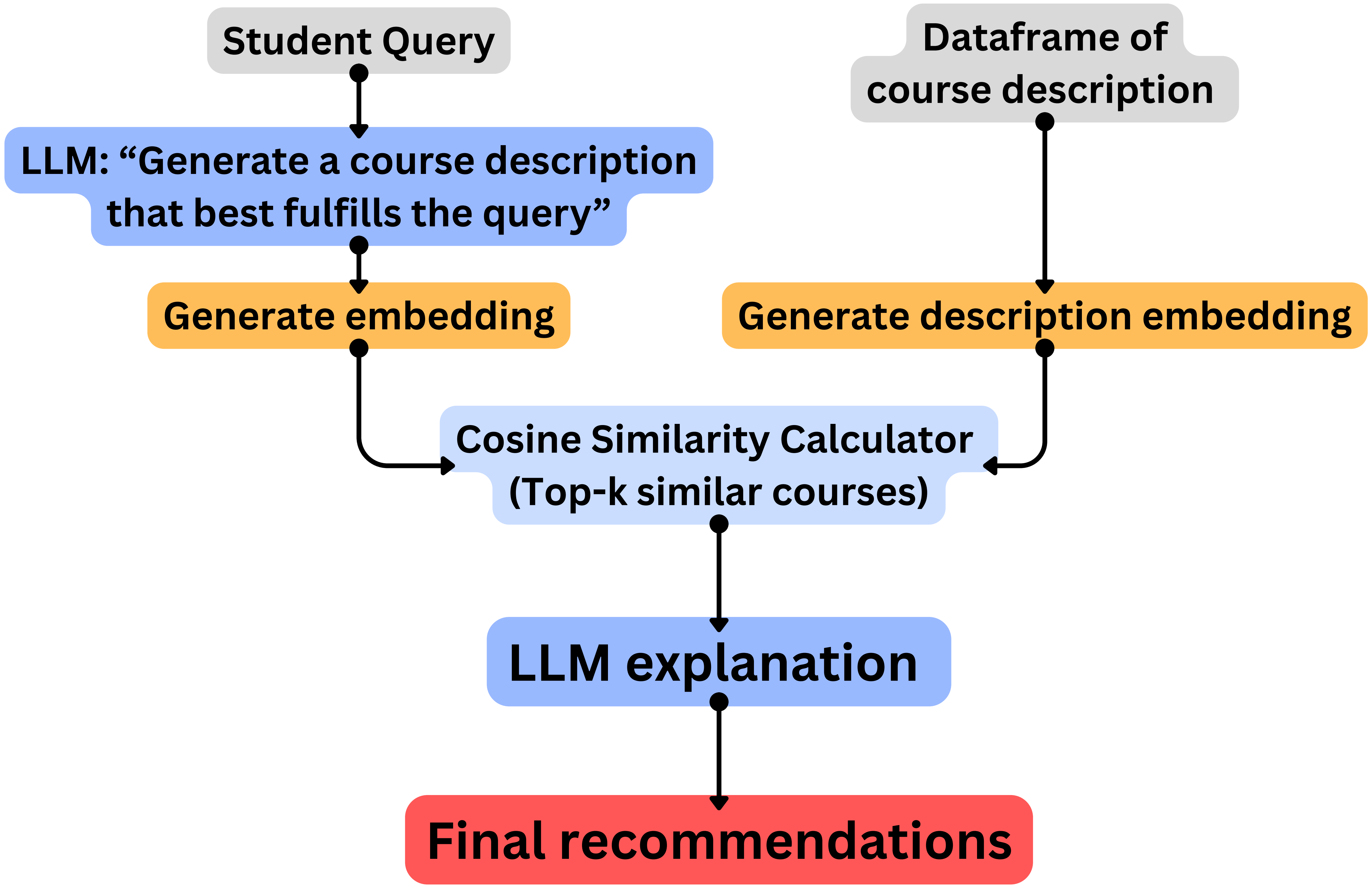
Title: "Applied Machine Learning and Deep Learning"

This comprehensive course delves into the practical application of Machine Learning (ML) and Deep Learning (DL) in real-world scenarios. The curriculum covers various topics including supervised, unsupervised, and reinforcement learning algorithms, with a strong focus on neural networks and deep learning architectures like Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM).

The course also explores essential concepts such as data preprocessing, model selection, optimization, evaluation metrics, and hyperparameter tuning. Students will gain hands-on experience through various projects that involve image recognition, natural language processing, speech recognition, and predictive analytics. By the end of this course, learners will have a solid understanding of machine learning and deep learning techniques to tackle real-world problems effectively.



# What is EduQuest?



# What is EduQuest?

**Student Query**

Eg. I'm interested in AI and how it can be applied in business decision-making. I want to learn both technical and practical aspects of machine learning

**Through embeddings + similarity search**

**Final recommendations**

1. IS4261: Designing AI-driven Business Innovations
Rationale: This course is ideal for students looking to apply AI and data-driven solutions to real-world business problems, fostering innovation in digital transformation.
Confidence: High

2. IS4400: Human-AI Interaction
Rationale: Given the increasing integration of AI in various industries, this course will equip students with essential skills in designing user-friendly and ethical AI systems that enhance human experience.
Confidence: High

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# DATA PREPROCESSING AND FILTERING

## Samples of Course Info AY25/26

| course | title | description | level | prefix | ori_course_code |
|---|---|---|---|---|---|
| CS2040 | Data Structure and Algorithms | This course introduces students to the design ... | 2000 | CS | CS2040 |
| MKT3427 (Cross-listed as MKT3722) | Research for Marketing Insights | Effective marketing research is necessary for ... | 3000 | MKT | MKT3427 |

**Cross-listed course merging rule:**
CS2040, CS2040C, CS2040S, CS2040DE → CS2040

🧠 If descriptions are  identical, courses are marked as cross-listed

⚙️ The model can filter courses by level and prefix effectively

# Workflow

| Step | LLM + Cosine Similarity Model |
|---|---|
| 1. Input | Student query |
| 2. Query Enrichment | LLM generates an **“example course description”** to expand meaning |
| 3. Query Embedding | Embed the generated example using ***nomic-embed-text*** |
| 4. Course Embeddings | Pre-computed & store in course[“embedding”] |
| 5. Similarity Computation | Calculate **cosine similarity** manually for every course and return **top 50 courses** |
| 6. Ranking and Selection | Recommender model **returns 10 courses**, re-rank by relevance and fit, and **writes short rationales** |
| 7. Explanation (LLM) | |
| 8. Output | Top 10 courses + explanations + confidence |

**Generator Model**

**Embedding Model**

**Recommender Model**

# MODELS WE USED

**Generator Model** mistral 7B

- Understand student queries in natural language
- Generate an “example course description” that captures what the student is looking for (e.g., “AI course combining business and machine learning”)

**Recommender Model** qwen2.5:7b-instruct 7B

- Explain the top recommended courses in human-readable form

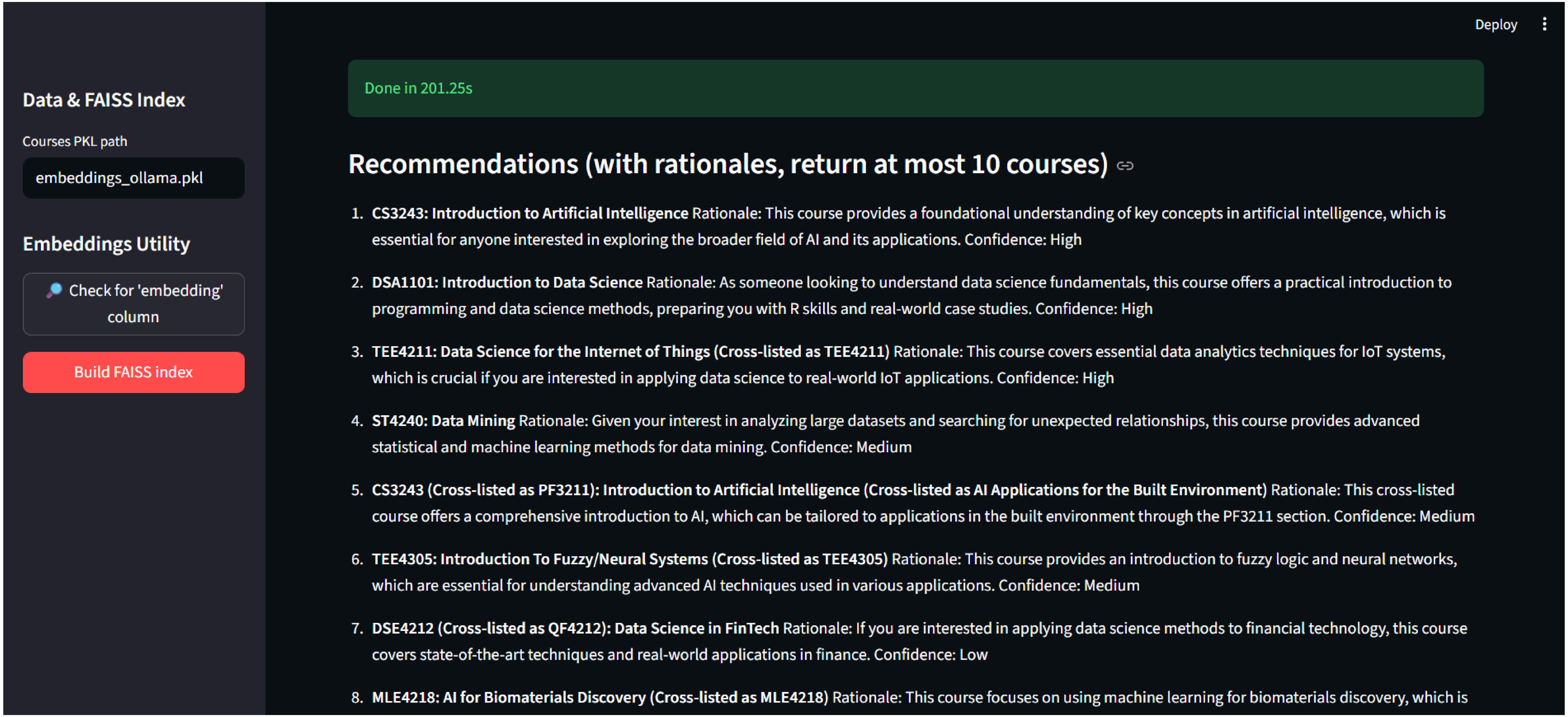
**Embedding Model** nomic-embed-text

- Converts text (course descriptions and query) into numerical vectors

# INTERFACE


Deploy
Data & FAISS Index
Courses PKL path
embeddings_ollama.pkl
Embeddings Utility
Check for 'embedding' column
Build FAISS index
EduQuest-NUS Recommender
Hybrid interface: LLM + Cosine vs FAISS | Type 0 = no rationales, Type 1 = with rationales
Search
Student query
I want to learn machine learning and deep learning for real-world applications.
Choose model
LLM + Cosine Similarity (recommend)
FAISS + Explanation (recommend_deterministic)
Filter by prefix
Choose options
Filter by level
1000
2000
3000
4000
Output Type
Without Rationales
Rationales

# OUTPUT


Deploy
Data & FAISS Index
Courses PKL path
embeddings_ollama.pkl
Embeddings Utility
Check for 'embedding' column
Build FAISS index
Done in 201.25s
Recommendations (with rationales, return at most 10 courses)
1. CS3243: Introduction to Artificial Intelligence Rationale: This course provides a foundational understanding of key concepts in artificial intelligence, which is essential for anyone interested in exploring the broader field of AI and its applications. Confidence: High
2. DSA1101: Introduction to Data Science Rationale: As someone looking to understand data science fundamentals, this course offers a practical introduction to programming and data science methods, preparing you with R skills and real-world case studies. Confidence: High
3. TEE4211: Data Science for the Internet of Things (Cross-listed as TEE4211) Rationale: This course covers essential data analytics techniques for IoT systems, which is crucial if you are interested in applying data science to real-world IoT applications. Confidence: High
4. ST4240: Data Mining Rationale: Given your interest in analyzing large datasets and searching for unexpected relationships, this course provides advanced statistical and machine learning methods for data mining. Confidence: Medium
5. CS3243 (Cross-listed as PF3211): Introduction to Artificial Intelligence (Cross-listed as AI Applications for the Built Environment) Rationale: This cross-listed course offers a comprehensive introduction to AI, which can be tailored to applications in the built environment through the PF3211 section. Confidence: Medium
6. TEE4305: Introduction To Fuzzy/Neural Systems (Cross-listed as TEE4305) Rationale: This course provides an introduction to fuzzy logic and neural networks, which are essential for understanding advanced AI techniques used in various applications. Confidence: Medium
7. DSE4212 (Cross-listed as QF4212): Data Science in FinTech Rationale: If you are interested in applying data science methods to financial technology, this course covers state-of-the-art techniques and real-world applications in finance. Confidence: Low
8. MLE4218: AI for Biomaterials Discovery (Cross-listed as MLE4218) Rationale: This course focuses on using machine learning for biomaterials discovery, which is

# Evaluation Setup

- **Dataset**: 25 queries paired with known relevant courses (AI, business, health, policy etc.)
- **Goal**: Check how many ground-truth courses appear in the model's Top-10 recommendations

## TestQueries

```
{
  "query": "I want to learn machine learning and deep learning for
                  real-world applications.",
  "relevant_courses": [
        "CG3201",
        "DSA5105",
        "CS3244"
   ]
}
```

## Model Evaluation

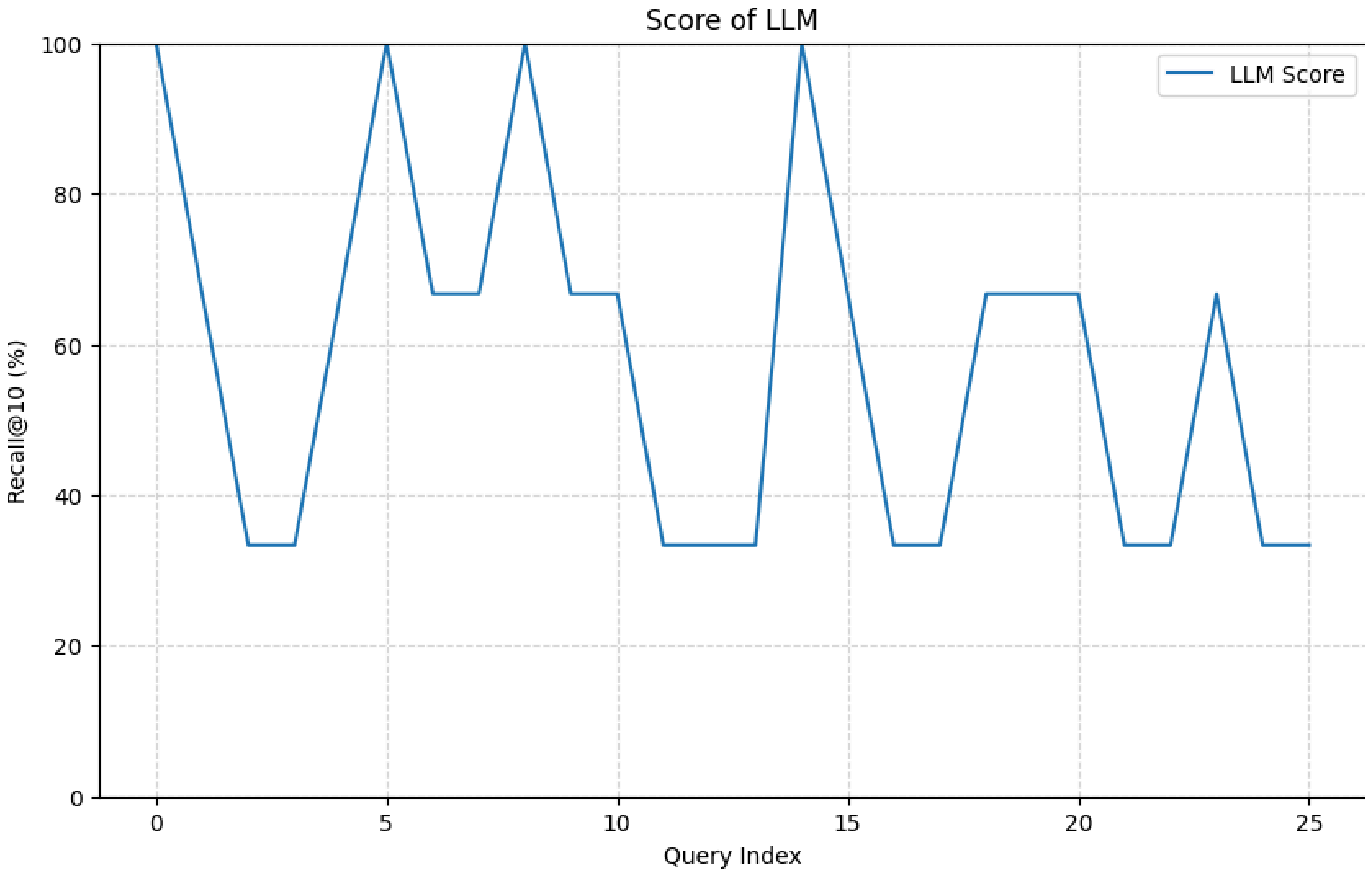
**Number of Test Queries relavant courses:** 3 material 1
**Top-10 courses recommended by model**: 10 material 2
**Metric**: Recall@10 = (# relevant found) / (total relevant)

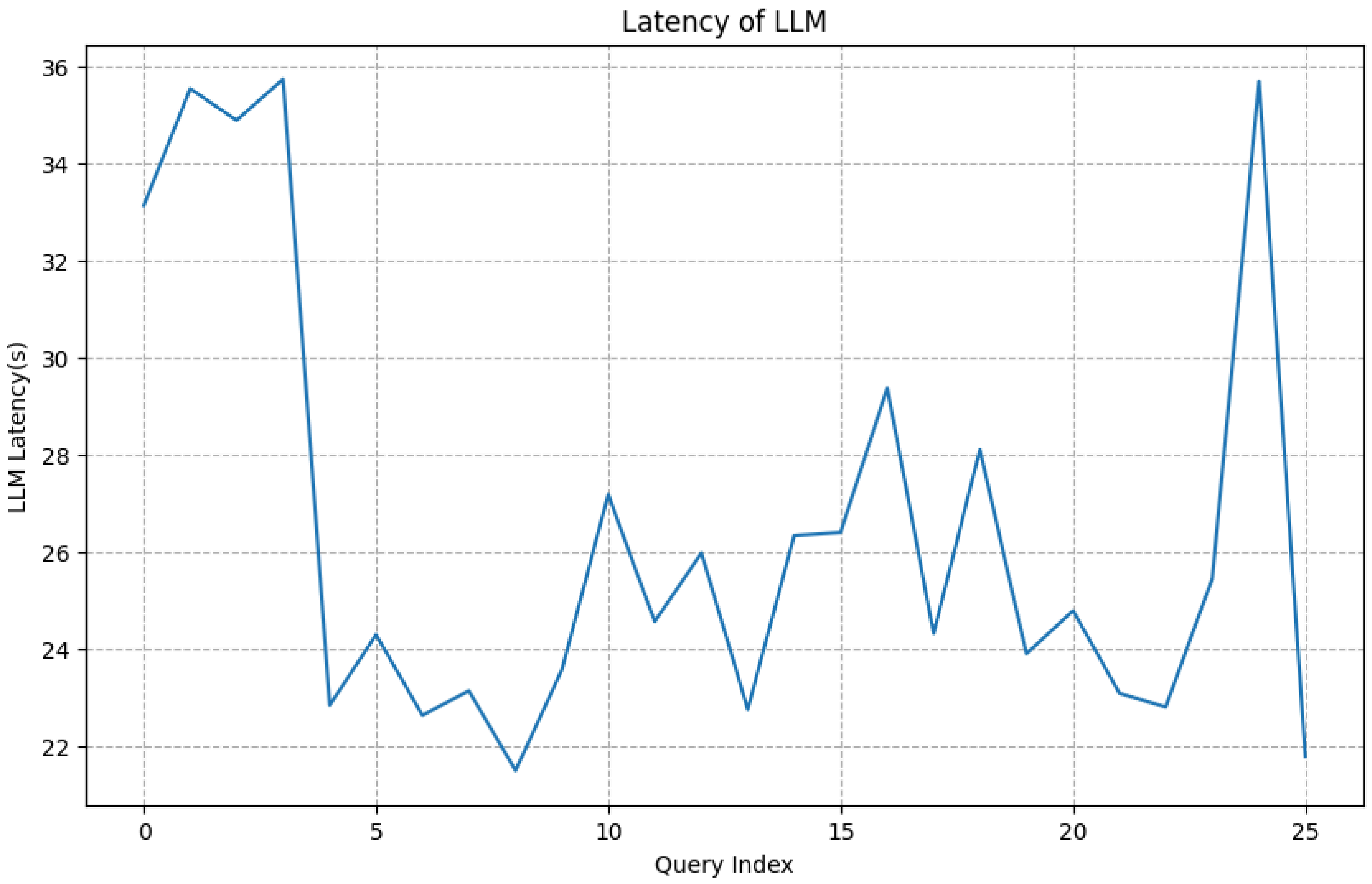
Example: 2 of 3 true relevant courses in top-10 → **Recall@10 = 0.67 (67%)**

# Evaluation Result

Average Recall@10 (Coverage): **57.69%**

Average latency: **26.54 s**




Proportion of relevant courses found among top-10 recommendations

# Strengths

- **High interpretability**
- **Context-rich matching**
- **Flexible**
- **Quality over speed**

# Limitations

- **Slow inference**
- **Inconsistent reproducibility**
- **Not scalable**

# Ideal Use-Cases

- When you need **high-quality, human-like recommendation**
- When queries are **ambiguous** or **abstract**
- For **low-volume** recommendation systems, not large scale search

# A FASTER OPTION?

Each query requires generating an example description and performance over **9000+ Cosine Similarity Calculations**, taking around **26 seconds per query**

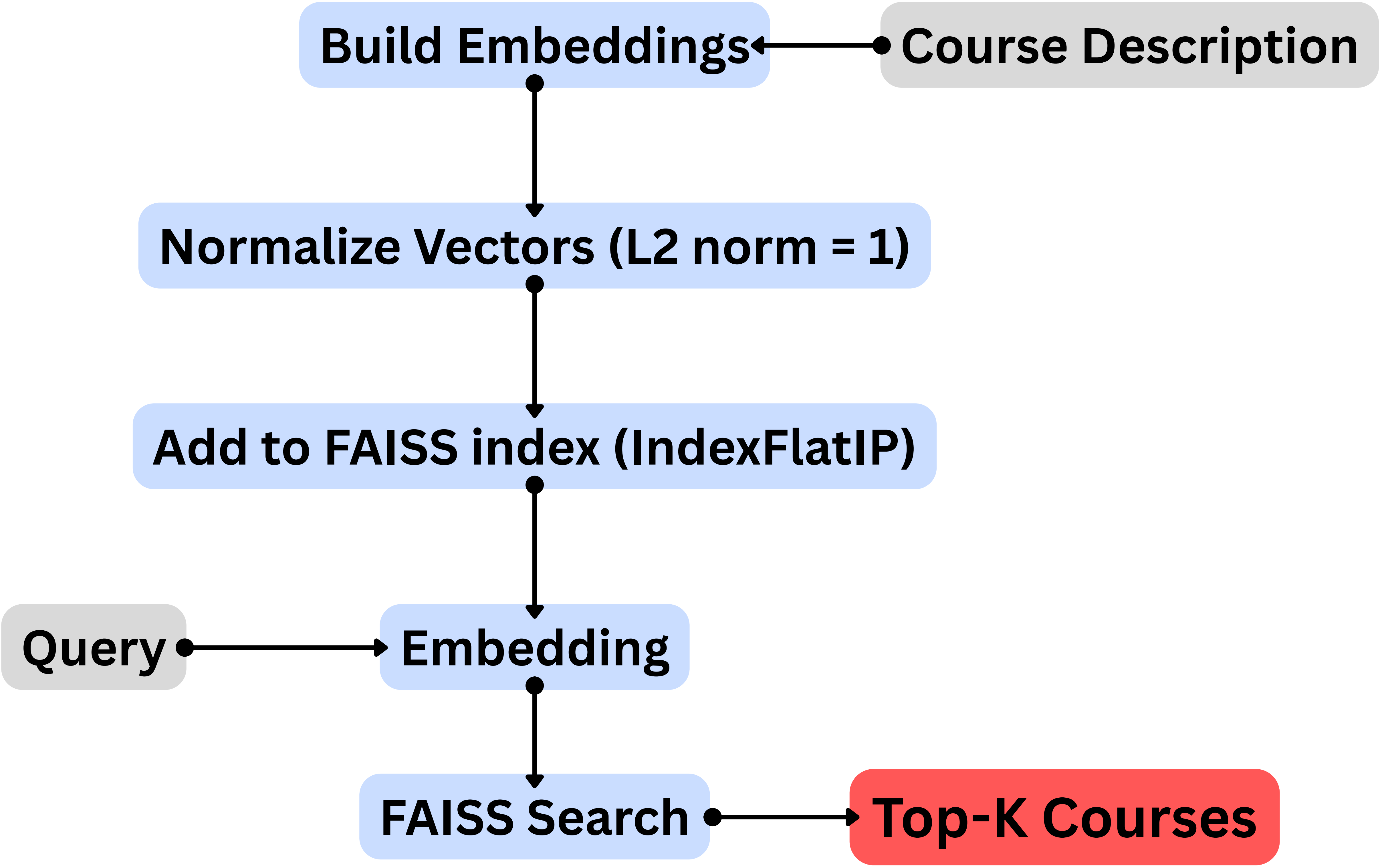
# Second Choice Model: FAISS

Facebook AI Similarity Search

- Cosine similarity gives accurate results but is computationally heavy.
- FAISS performs the same semantic comparison — but much faster and more scalable.

Meta

# How FAISS works?



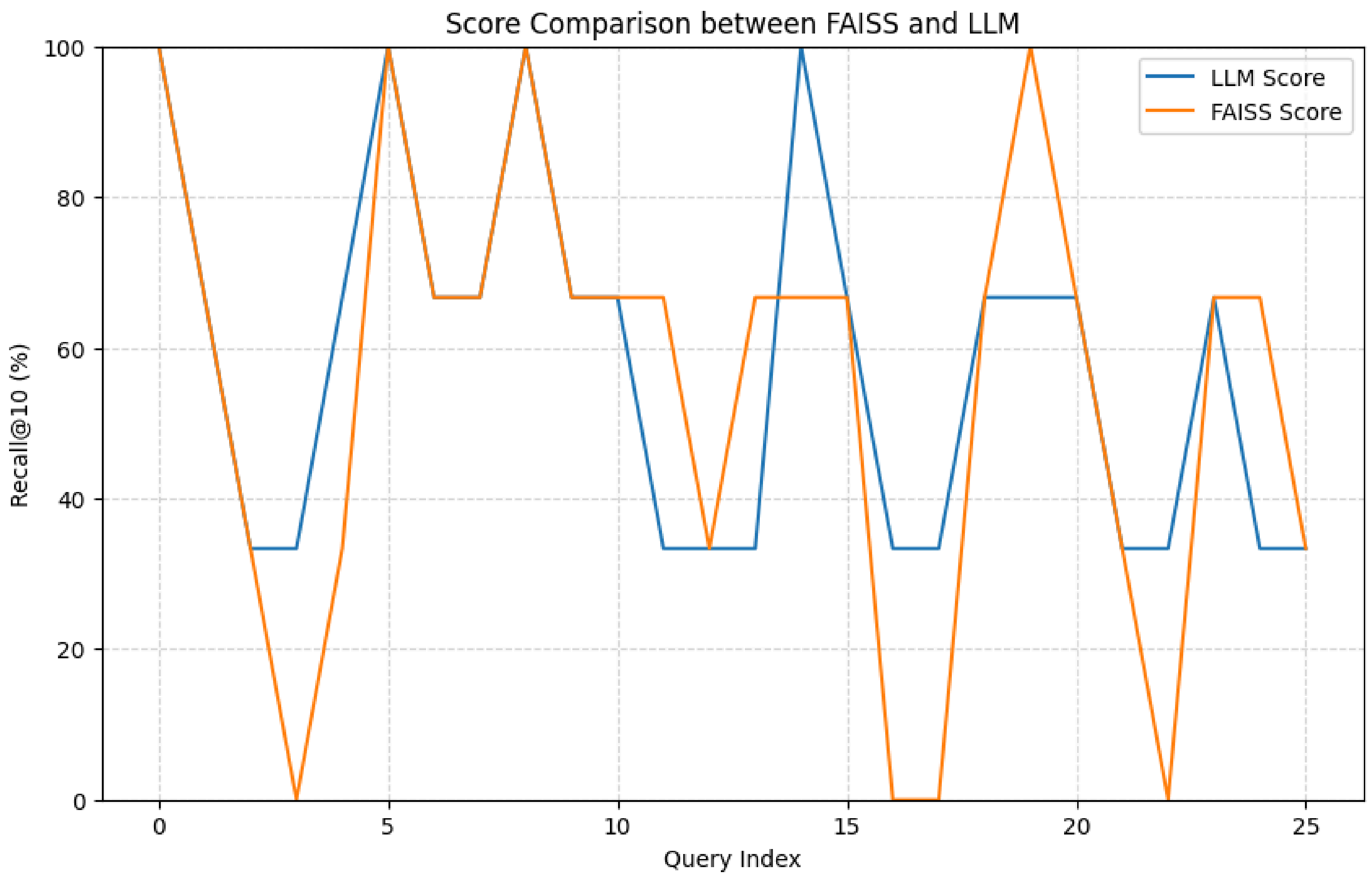
# Workflow Comparison: LLM + Cosine Similarity vs FAISS

| Step | LLM + Cosine Similarity Model | FAISS Model | Key Difference |
|---|---|---|---|
| 1. Input | Student query | Same query text | |
| 2. Query Enrichment | LLM generates an **“example course description”** to expand meaning | **No generation step; use query as-is** | **LLM version understands nuance but add latency** |
| 3. Query Embedding | Embed the generated example using ***nomic-embed-text*** | Embed the raw query using ***nomic-embed-text*** | Both produce vector representations |
| 4. Course Embeddings | Pre-computed & store in course[“embedding”] | Pre-computed & stored in FAISS index | |
| 5. Similarity Computation | Calculate **cosine similarity** manually for every course and return **top 50 courses** | **FAISS IndexFlatIP** performs fast inner-product search and return **top 10 courses** | |
| 6. Ranking and Selection | Recommender model **returns 10 courses**, re-rank by relevance and fit, and **writes short rationales** | **FAISS returns top K directly** and similarity scores | |
| 7. Explanation (LLM) | | **Optional** - Explain why each of the courses fits the student’s interest | Only LLM version gives human-readable reasons |
| 8. Output | Top 10 courses + explanations + confidence | Top 10 courses + similarity scores + (optional) explanations | **FAISS focuses on retrieval speed** |

# Evaluation Result

**Average Recall@10 (Coverage):**

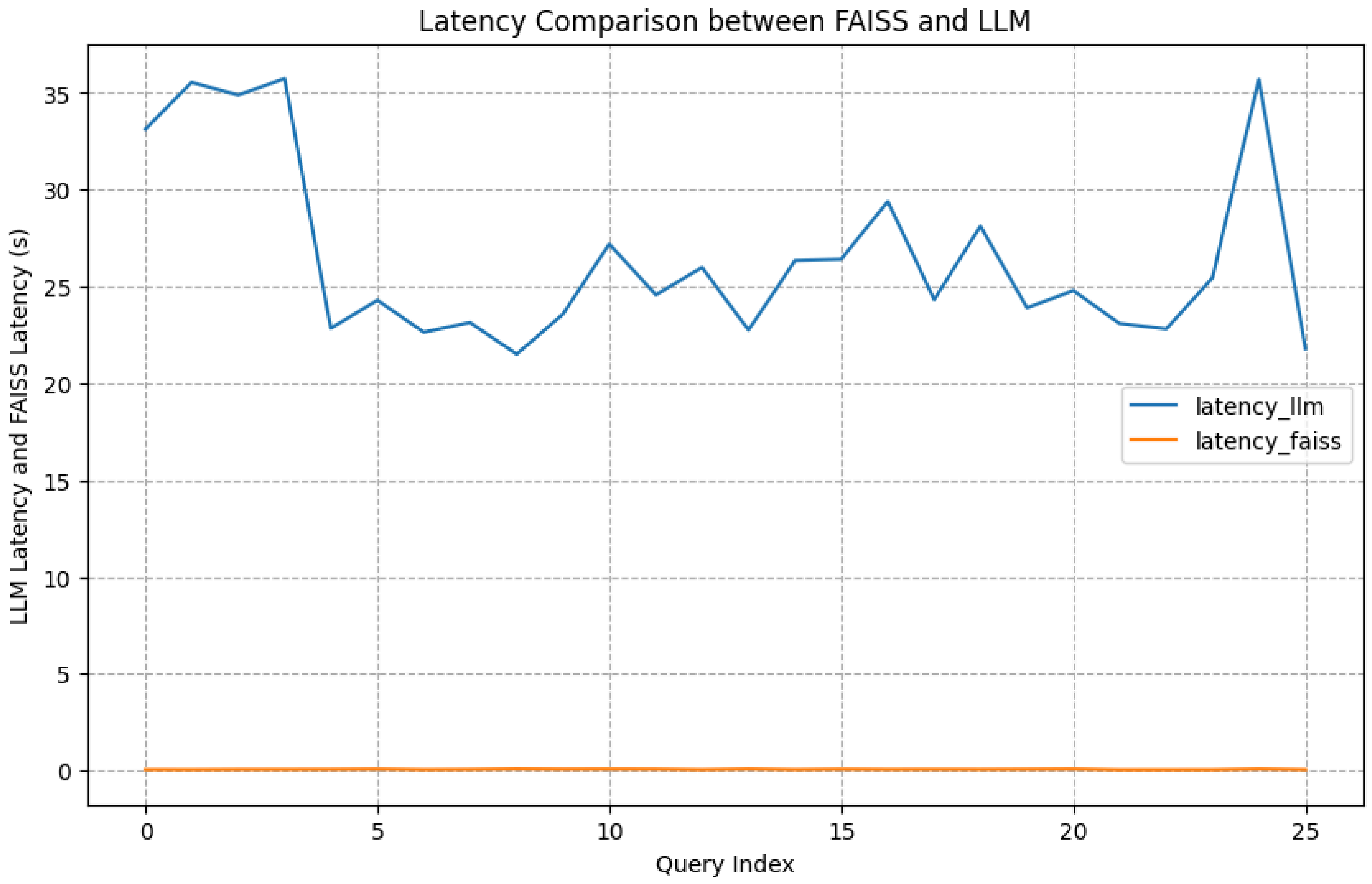
- **LLM Model**: 57.69%
- **FAISS Model**: 55.13%




Recall@10 comparison per query (LLM vs FAISS)

**Average Latency:**

- **LLM Model**: 26.54 seconds
- **FAISS Model**: 0.07 seconds




Latency comparison per query (LLM vs FAISS)

# Comparing Two Recommendation Models: LLM + Cosine Similarity vs FAISS

| LLM + Cosine Similarity | FAISS Recommender |
|---|---|
| Concept / Flow<br>• LLM expand query<br>• Cosine sim = semantic match | Concept / Flow<br>• Embedding index built<br>• Fast vector search |
| ✅ Pro: High quality, human-like<br>❌ Cons: Slow, inconsistent | ✅ Pros: Fast, scalable<br>❌ Cons: Less nuanced |

# THANK YOU FOR LISTENING!

Presentation Slides

Github Repo